UNIVERSITY OF TEHRAN
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

Computer Assignment 3: Network Emulation
GNS3, VLANs, and Routing Protocols

Course: Computer Networks
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1. Emulation Environment Setup

The emulation environment was configured using GNS3. Cisco `c7200-adventerprisek9-mz` router images were imported and allocated specific hardware slots to ensure adequate port availability:

- **Slot 1 (PA-2FE-TX):** Adds two extra Fast Ethernet ports to the router. It is used to connect the router to local devices like switches or PCs.
- **Slot 2 (PA-4T+):** Adds four serial ports to the router, used specifically for WAN connections to simulate long-distance links between different networks.

2. Basic Switching Topology

2.1 Objective and Architecture

A hierarchical switching topology was implemented consisting of a distribution switch (Switch 3) and two access switches (Switch 1 and Switch 2). Four Virtual PCs were assigned IP addresses within the `192.168.1.0/24` broadcast domain.

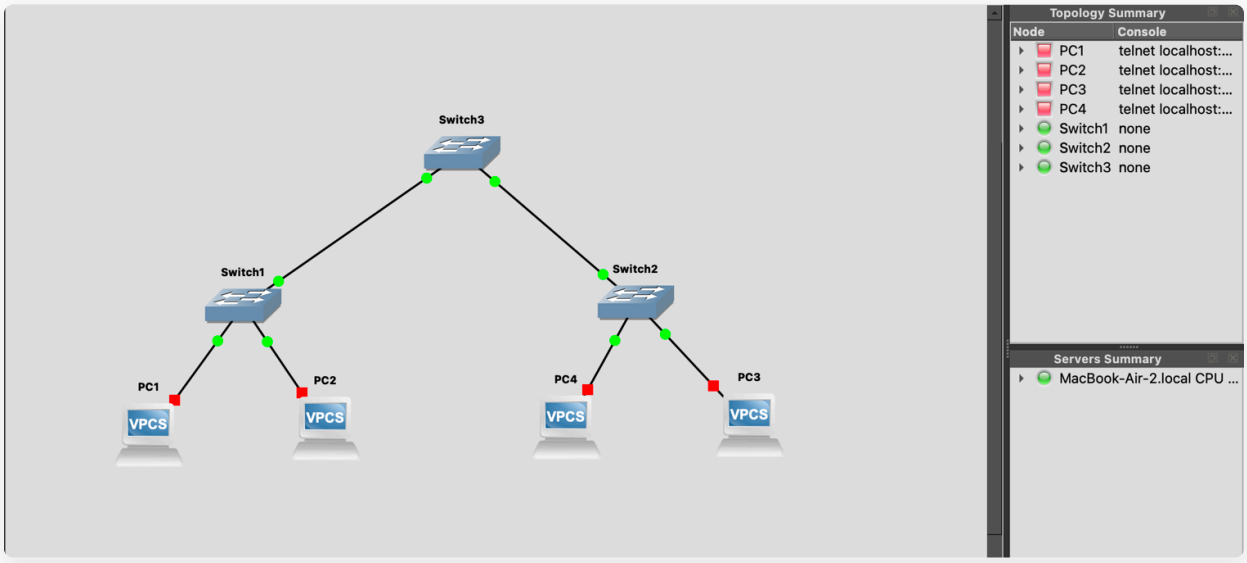


Figure 2.1: *GNS3 workspace displaying the hierarchical switching topology.*

2.2 Results and Verification

> **Analysis:** When the simulation first starts, the ARP tables of the VPCSs are empty. Using the `ping` command in their console, simple ICMP messages are exchanged so that the ARP table can be dynamically updated.

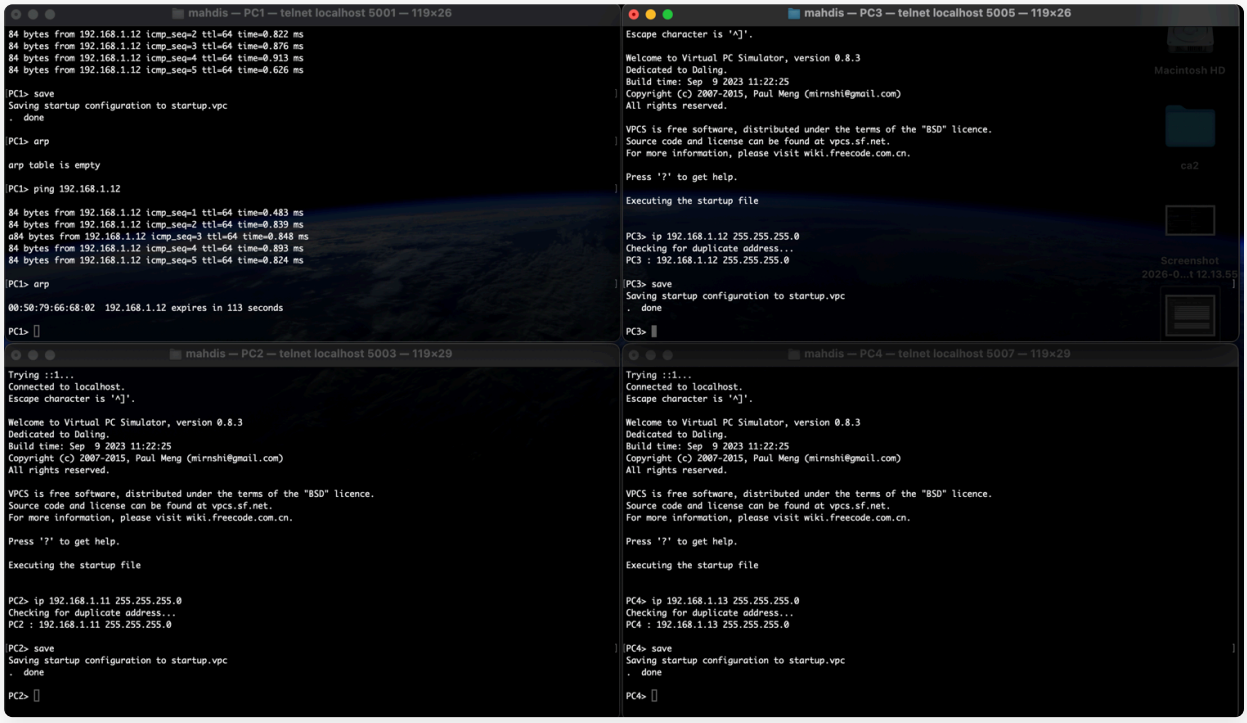


Figure 2.2: *Successful ICMP communication and ARP table generation between PCs across the multi-switch fabric.*

3. Virtual Local Area Networks (VLANs) & Inter-VLAN Routing

3.1 Network Topology

To enable communication between VLAN 10 and VLAN 20, a Router-on-a-Stick topology was deployed. The switch port connected to Router 1 (R1) was configured as an 802.1Q trunk.

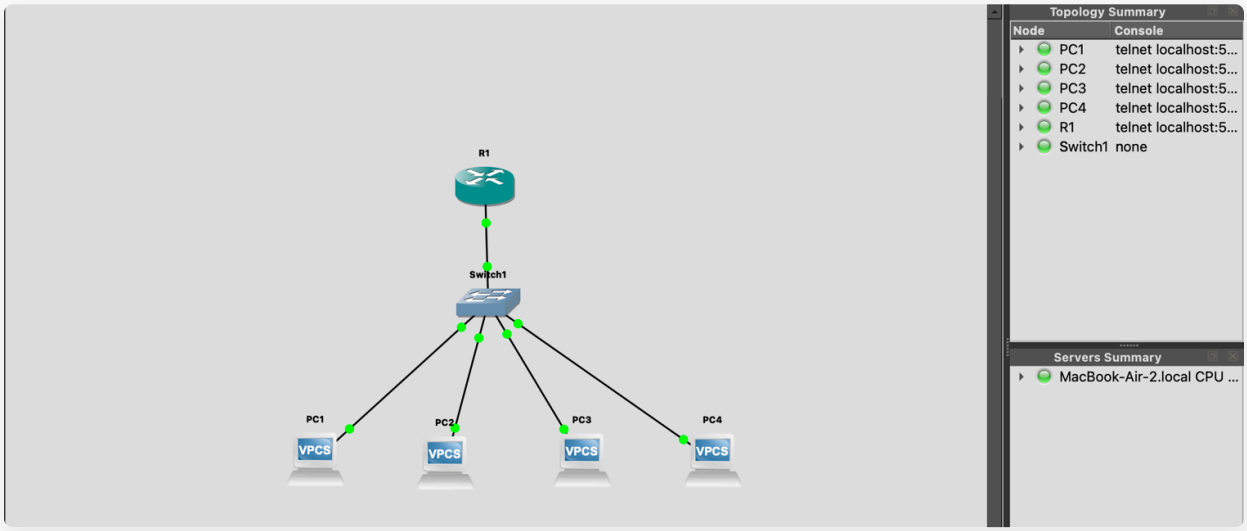


Figure 3.1: *GNS3 workspace displaying the Router-on-a-Stick VLAN topology.*

3.2 Architecture and Router Configuration

On R1, the physical interface was divided into two logical sub-interfaces (`f0/0.10` and `f0/0.20`), acting as the default gateways for their respective VLANs.

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#interface FastEthernet0/0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#interface FastEthernet0/0.10
R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.0
R1(config-subif)#exit
R1(config)#
R1(config)#interface FastEthernet0/0.20
R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip address 192.168.20.1 255.255.255.0
R1(config-subif)#end
R1#
R1#write memory
*Jun 17 07:12:41.839: %SYS-5-CONFIG_I: Configured from console by console
R1#write memory
*Jun 17 07:12:42.863: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Jun 17 07:12:43.863: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1#write memory
```

Figure 3.2: *Configuration of 802.1Q encapsulation and IP addressing on R1's sub-interfaces.*

3.3 Routing Table Verification

> **Analysis:** By using the `show ip route` command, we can verify that the router has successfully identified both VLAN subnets as "directly connected" (denoted by the 'C'). This indicates that R1 is now capable of routing traffic between the two separate broadcast domains.

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.10.0/24 is directly connected, FastEthernet0/0.10
L        192.168.10.1/32 is directly connected, FastEthernet0/0.10
      192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.20.0/24 is directly connected, FastEthernet0/0.20
L        192.168.20.1/32 is directly connected, FastEthernet0/0.20
R1#
```

Figure 3.3: *R1 Routing Table confirming the successful attachment of VLAN 10 and VLAN 20.*

3.4 Inter-VLAN Connectivity Results

With the router fully configured, connectivity was tested across the network. The initial ICMP Echo Requests experienced a slight delay as the router utilized ARP to resolve the destination MAC addresses. Subsequent packets were routed seamlessly with minimal latency, proving successful Inter-VLAN communication.

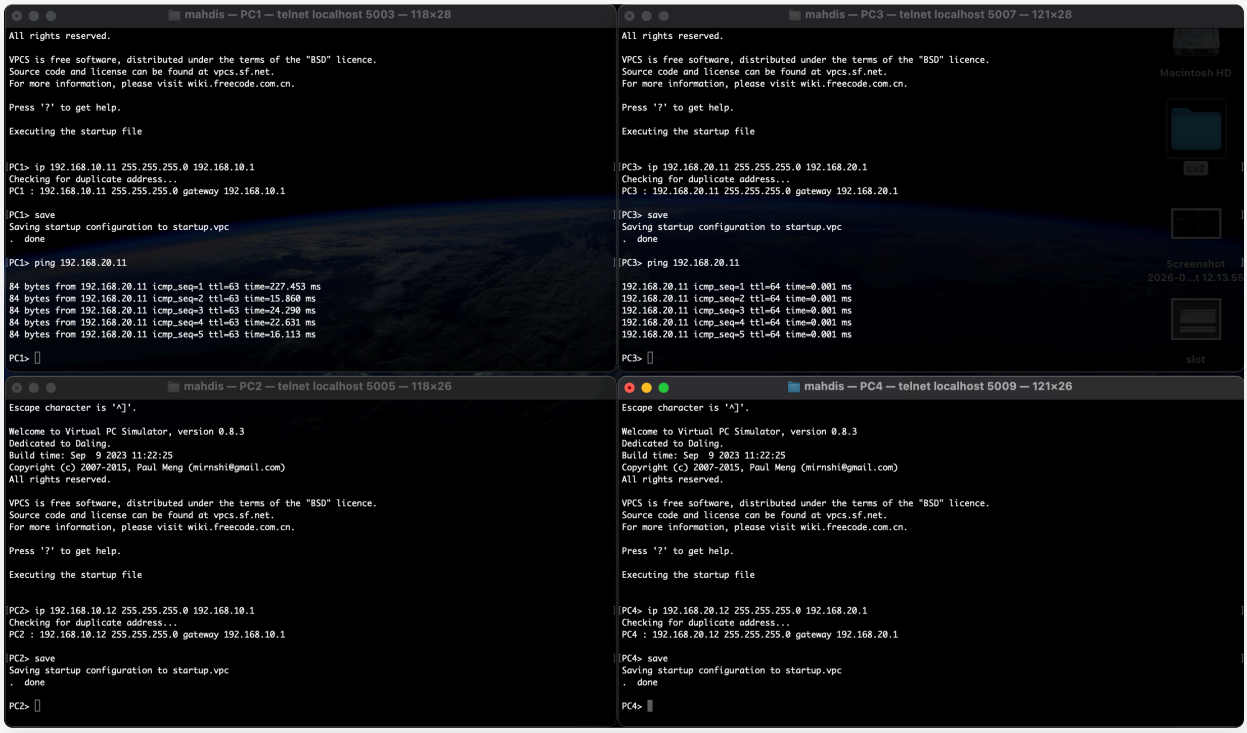


Figure 3.4: *Combined console view of PC1 through PC4 displaying IP assignments, gateway configurations, and successful inter-VLAN ping results.*

4. Static Routing & Link Redundancy

4.1 Network Topology

A four-router core network was established using serial point-to-point links operating on `/30` subnets to simulate a Wide Area Network (WAN). Edge networks were connected via FastEthernet interfaces.

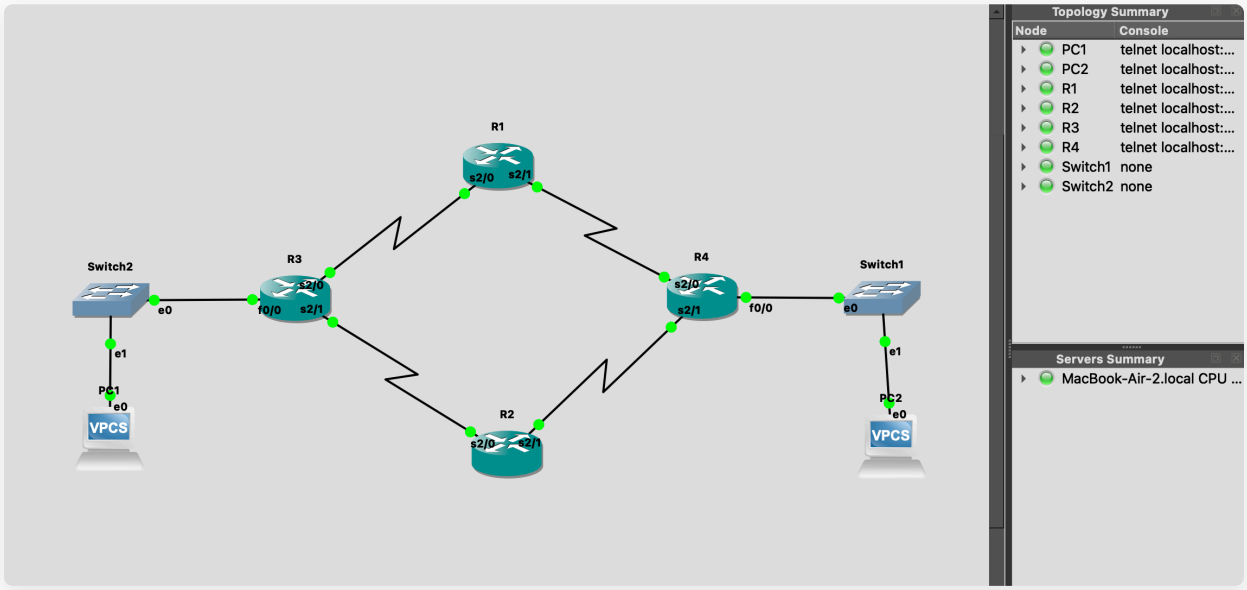


Figure 4.1: *GNS3 workspace displaying the 4-router WAN topology with redundant serial links.*

4.2 Static Route Configuration

Static routes were explicitly defined on all routers to ensure end-to-end connectivity. To provide fault tolerance, "Floating Static Routes" were configured with a higher Administrative Distance (AD = 10) to act as backup paths.

```
! Primary and Backup route configuration on R3
R3(config)# ip route 192.168.4.0 255.255.255.0 10.0.0.2
R3(config)# ip route 192.168.4.0 255.255.255.0 10.0.0.10 10
```

4.3 Normal Routing State

Initial connectivity was verified via ICMP. The routing tables for all four routers successfully registered the statically assigned primary paths.

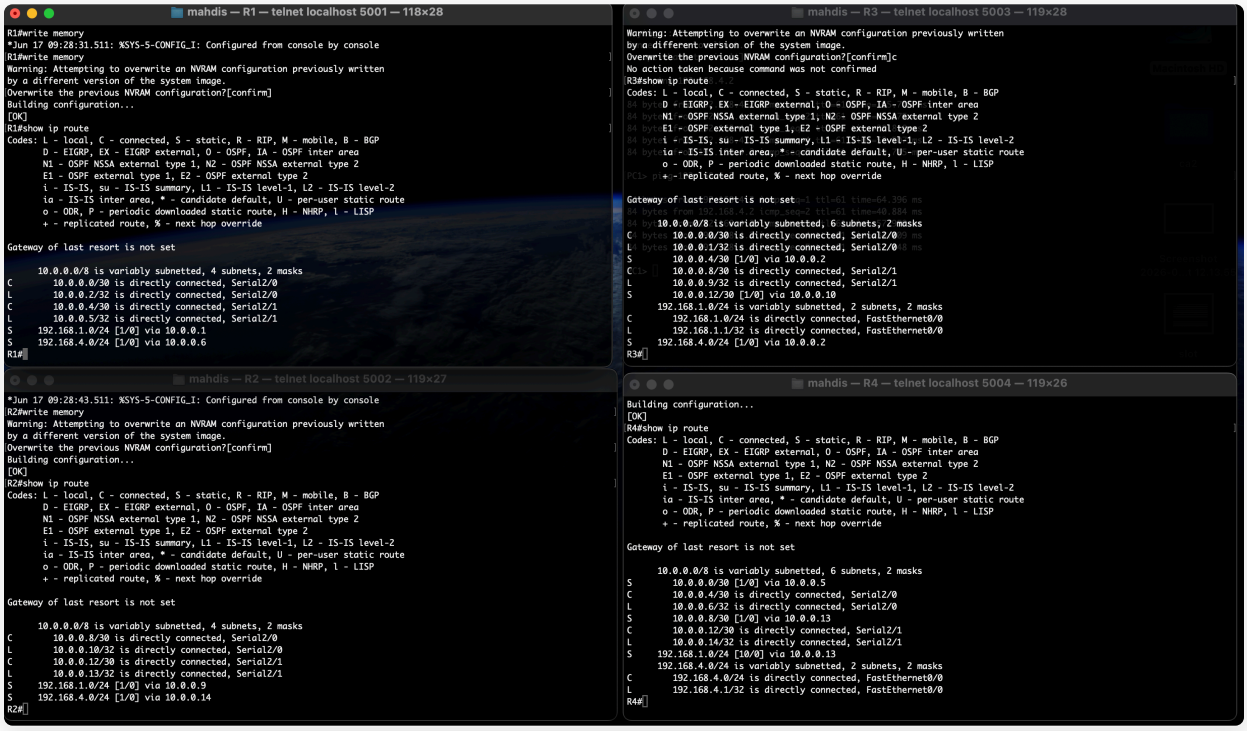


Figure 4.2: *Normal state routing tables across the four core routers.*

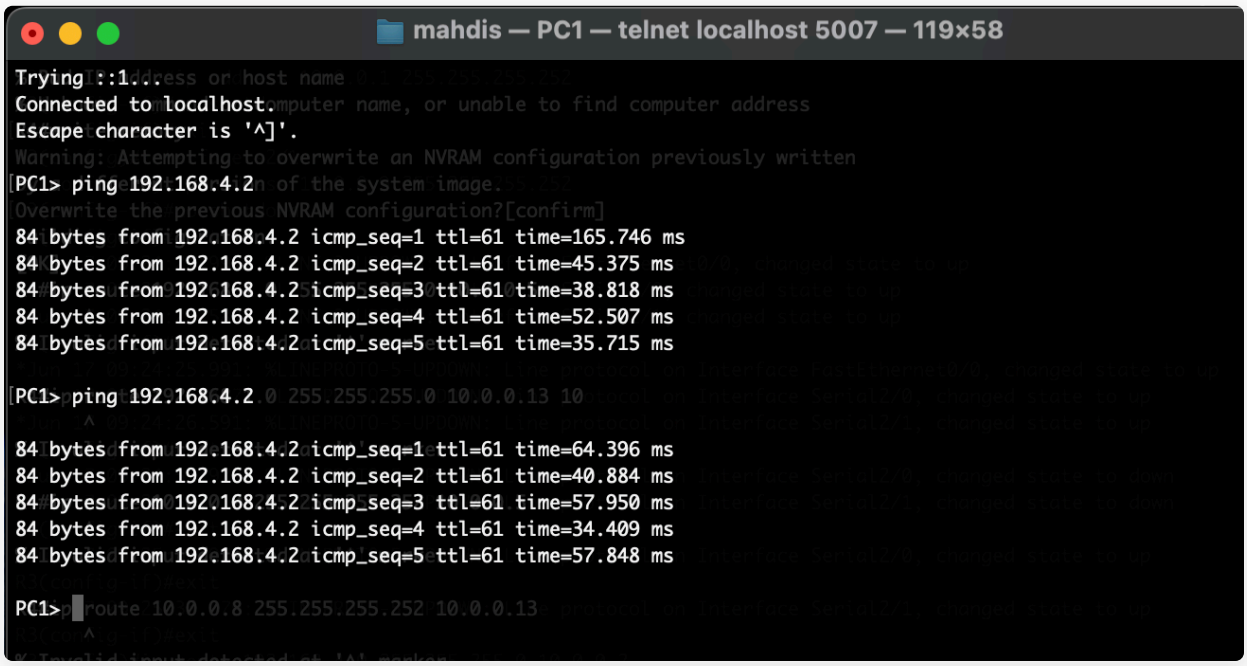
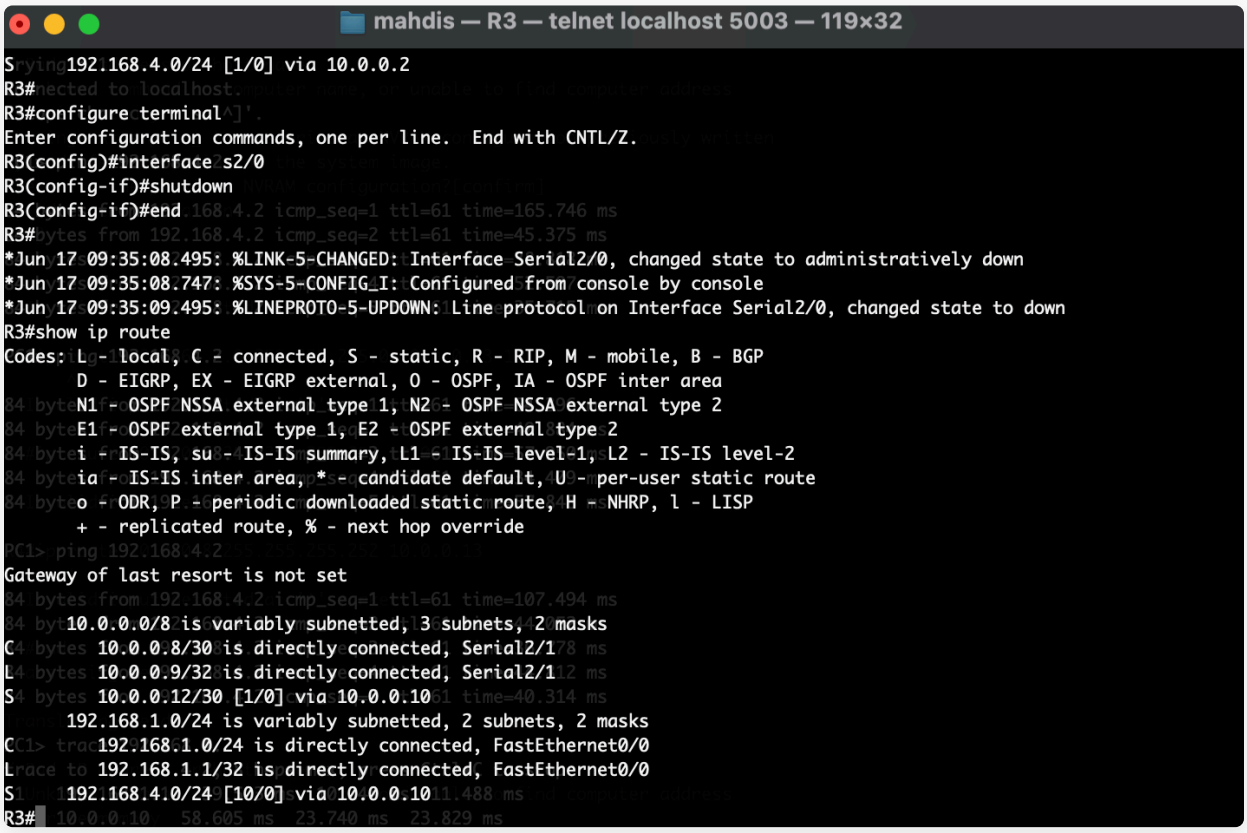


Figure 4.3: *Successful end-to-end ICMP ping under normal network conditions.*

4.4 Failover Simulation

To test the redundant architecture, the primary serial interface (`s2/0` on R3) was manually shut down. The routing table immediately converged to utilize the floating static route (via R2), maintaining uninterrupted connectivity between the edge networks.



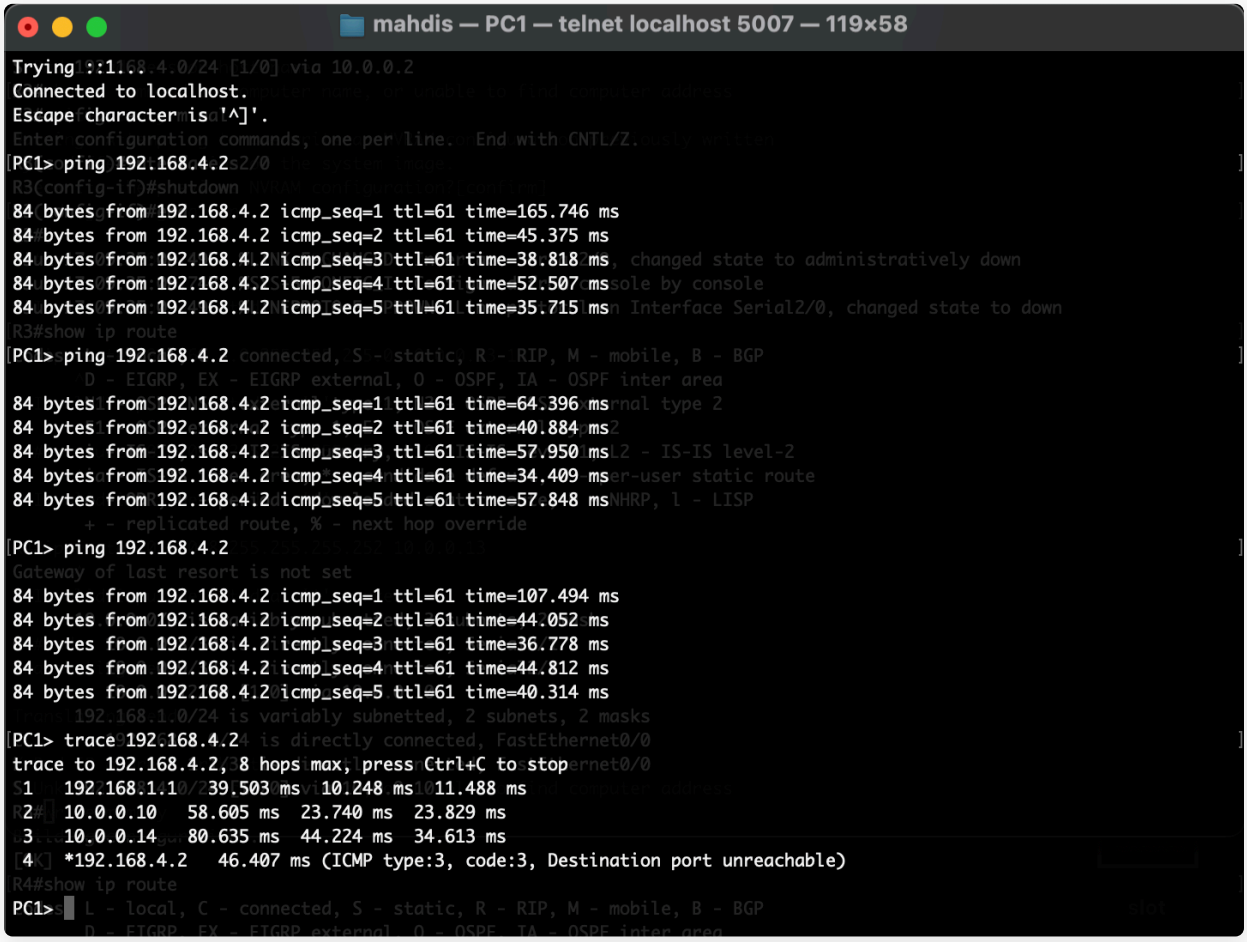


Figure 4.4: Left: Updated routing table on R3 showing the activated backup route. Right: Sustained ping and trace route verifying the new path through 10.0.0.10.

5. Dynamic Routing (OSPF)

5.1 Network Topology Setup

To eliminate the scalability limitations of static routing, the network was transitioned to use the Open Shortest Path First (OSPF) Link-State routing protocol. The IP addressing scheme was updated to reflect the new dynamic routing scenario.

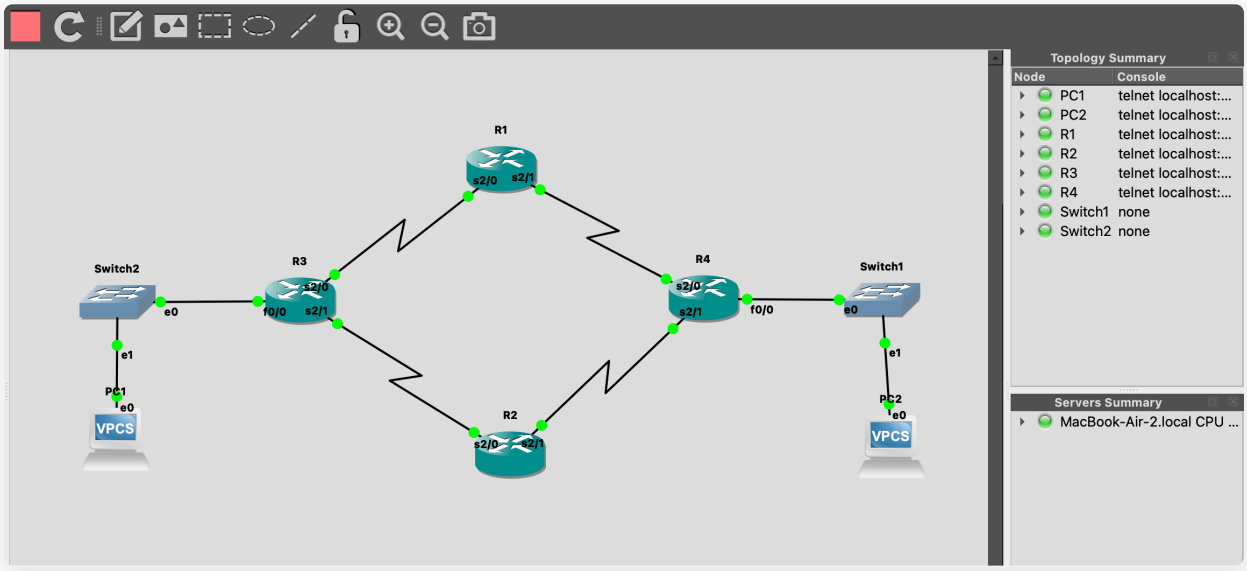


Figure 5.1: *Updated GNS3 topology prepared for OSPF dynamic routing.*

5.2 OSPF Configuration

Static routes were removed from the routers to prevent conflicts. OSPF Process ID 1 was then enabled globally on all core routers, and connected networks were advertised into Area 0 (the backbone area) using wildcard masks.

```
! Example OSPF Configuration on R3
R3(config)# router ospf 1
R3(config-router)# network 192.168.1.0 0.0.0.255 area 0
R3(config-router)# network 10.0.0.0 0.0.0.3 area 0
R3(config-router)# network 10.0.0.8 0.0.0.3 area 0
```

5.3 Verification and Network Convergence

Once configured, the routers successfully exchanged Link-State Advertisements (LSAs), formed neighbor adjacencies, and populated their routing tables automatically. The final end-to-end ping test was completely successful, confirming that OSPF had correctly mapped the topology. The initial ping delay reflects the time

required for the network to resolve addresses dynamically across the newly established OSPF routes.

```
R3#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
10.0.0.13         0   FULL/  -        00:00:39    10.0.0.10       Serial2/1
10.0.0.5          0   FULL/  -        00:00:39    10.0.0.2        Serial2/0
```

```
R3#show ip ospf database

            OSPF Router with ID (192.168.1.1) (Process ID 1)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
10.0.0.5        10.0.0.5        1362        0x80000004 0x00D20B 4
10.0.0.13       10.0.0.13       1367        0x80000003 0x001896 4
192.168.1.1     192.168.1.1     1362        0x80000003 0x0074E2 5
192.168.4.1     192.168.4.1     580         0x80000003 0x00FF3C 5
R3#
```

mahdis — R3 — telnet localhost 5003 — 119×29

```
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.0.0.13         0   FULL/  -        00:00:39    10.0.0.10       Serial2/1
10.0.0.5          0   FULL/  -        00:00:39    10.0.0.2        Serial2/0
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C        10.0.0.0/30 is directly connected, Serial2/0
L        10.0.0.1/32 is directly connected, Serial2/0
O        10.0.0.4/30 [110/128] via 10.0.0.2, 00:01:23, Serial2/0
C        10.0.0.8/30 is directly connected, Serial2/1
L        10.0.0.9/32 is directly connected, Serial2/1
O        10.0.0.12/30 [110/128] via 10.0.0.10, 00:01:23, Serial2/1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, FastEthernet0/0
L        192.168.1.1/32 is directly connected, FastEthernet0/0
O     192.168.4.0/24 [110/129] via 10.0.0.10, 00:01:23, Serial2/1
                     [110/129] via 10.0.0.2, 00:01:23, Serial2/0
R3#
```

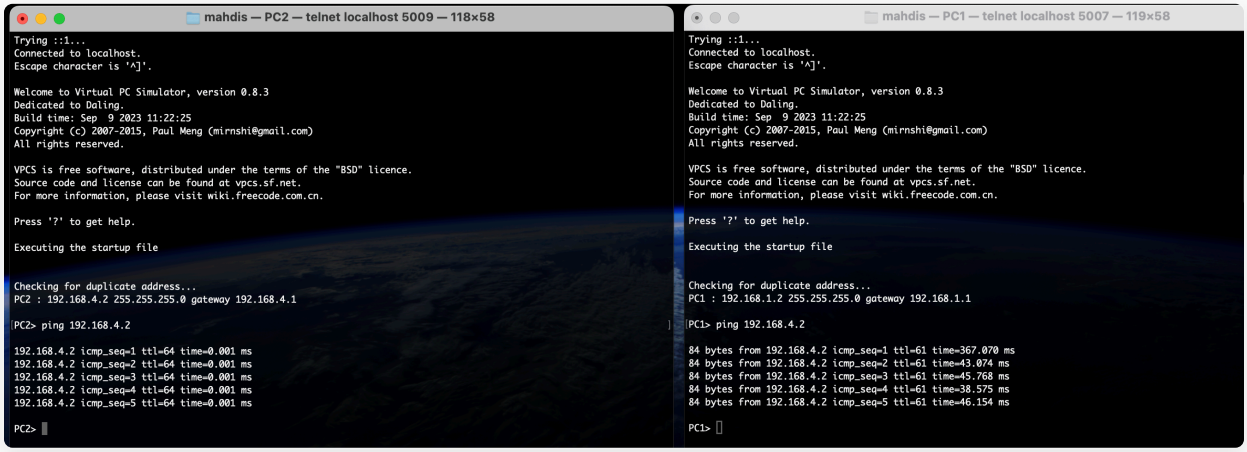


Figure 5.2: OSPF Diagnostic outputs. Top Left: Neighbor adjacencies in the FULL state. Top Right: Link-State Database sharing router IDs. Bottom Left: Dynamically learned paths identified by 'O' routes. Bottom Right: End-to-end ping verification showing successful routing.

6. Conclusion

This computer assignment successfully demonstrated the progression of network design from simple Layer 2 switching to complex Layer 3 routing. By implementing inter-VLAN segmentation, manually configuring static routes with failover redundancy, and finally deploying the OSPF dynamic routing protocol, we verified the fundamental mechanisms that allow robust, fault-tolerant communication across Wide Area Networks.